THE DUALIZING MODULE OF A CONNECTED NOETHERIAN SCHEME IS (S_2) WITH SUPPORT ON THE MAXIMAL DIMENSIONAL COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AWK (sources/stacks-duality/pinned-original-source.tex line 6101).

Allowed standard prerequisites (editorial).

- Uniqueness of a dimension function up to a constant on a connected scheme. Topology, Lemma labelled topology-lemma-dimension-function-unique, cited at proof lines 6128-6130. Allows choosing the dimension function of the dualizing complex.
- The dimension function attached to a dualizing complex. Same chapter, Lemma labelled lemma-dimension-function-scheme, cited at proof lines 6130-6132. Supplies the function used.
- The algebraic depth and support statement for the dualizing module. Dualizing Complexes, Lemma labelled dualizing-lemma-depth-dualizing-module, cited at proof lines 6134-6136, with Cohomology of Schemes, Definition labelled coherent-definition-depth, cited at proof lines 6136-6140. Gives both assertions.

Source reference. Original lemma 02IB: sources/stacks-chow/tags.txt, source lines 3294–3294. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6113 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWF: sources/stacks-duality/pinned-original-source.tex, source lines 567–567. Same-chapter pointer into the author environment lemma at source line 567; invoked at proof line 6116 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0AWE: sources/stacks-chow/tags.txt, source lines 13717–13717. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6119 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 0341: sources/stacks-chow/tags.txt, source lines 4076–4076. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6121 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a connected Noetherian scheme and let ω_X be a dualizing module on X. The support of ω_X is the union of the irreducible components of maximal dimension with respect to any dimension function and ω_X is a coherent $\mathcal{O}_X$-module having property (S_2).*

Proof. By our conventions discussed above there exists a dualizing complex $\omega_X^\bullet$ such that ω_X is the leftmost nonvanishing cohomology sheaf. Since X is connected, any two dimension functions differ by a constant (Topology, Lemma 02IB). Hence we may use the dimension function associated to $\omega_X^\bullet$ (Lemma 0AWF). With these remarks in place, the lemma now follows from Dualizing Complexes, Lemma 0AWE and the definitions (in particular Cohomology of Schemes, Definition 0341). □